

Table 2. False-premise correction and normal-question preservation

Method	Qwen2.5 7B				GPT-6 Luna				Gemma 4 12B				Qwen3.8 27B OFF				Qwen3.8 27B ON			
	Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE		Cancer-Myth		CREPE	
	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑	FPQ ↑	NFP ↑	FPQ ↑	TPQ ↑
TF-IDF text gate	50.7 ^H	60.4 ^H	—	—	—	—	—	—	63.8 ^H	50.3 ^H	—	—	68.3 ^H	49.0 ^H	—	—	78.2 ^H	48.3 ^H	—	—
Plain	2.4 ^H	100.0 ^H	—	—	54.5 ^L	91.0 ^L	72.7 ^R	98.9 ^R	4.3 ^H	100.0 ^H	—	—	49.7 ^H	79.9 ^H	—	—	40.7 ^H	88.6 ^H	—	—
CoT	0.7 ^H	99.3 ^H	—	—	—	—	—	—	3.4 ^H	99.3 ^H	—	—	54.7 ^H	71.1 ^H	—	—	43.1 ^H	87.9 ^H	—	—
Balanced instruction	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Premise-review CoT (2-step)	10.5 ^H	94.6 ^H	—	—	—	—	—	—	41.2 ^H	78.5 ^H	—	—	61.4 ^H	55.0 ^H	—	—	66.7 ^H	57.0 ^H	—	—
Always correct	56.9 ^H	21.5 ^H	—	—	—	—	—	—	72.0 ^H	2.0 ^H	—	—	71.0 ^H	28.9 ^H	—	—	83.4 ^H	5.4 ^H	—	—
PreWoMe	—	—	—	—	88.9 ^L	59.0 ^L	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Extract + Verify	—	—	—	—	87.9 ^L	48.0 ^L	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Direct-gated response	6.9 ^H	90.6 ^H	—	—	—	—	—	—	37.2 ^H	83.9 ^H	—	—	53.0 ^H	72.5 ^H	—	—	53.5 ^H	75.8 ^H	—	—
Premise-review gated response	21.3 ^H	85.9 ^H	—	—	—	—	—	—	43.4 ^H	67.1 ^H	—	—	62.1 ^H	59.7 ^H	—	—	62.1 ^H	60.4 ^H	—	—
Probe-gated response	44.8 ^H	64.4 ^H	—	—	n/a	n/a	n/a	n/a	—	—	—	—	—	—	—	—	—	—	—	—
GEPA	—	—	—	—	78.8 ^L	72.0 ^L	74.7 ^R	98.0 ^R	—	—	—	—	—	—	—	—	—	—	—	—
Ours	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Stored results; —: not measured/comparable; n/a: outside the defined implementation. Percentages, one run.
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.
FPQ / NFP / TPQ: Well ≥ 4 (%). Medical and general QA preservation is reported in Table 3.
L: Luna matched 99 FPQ / 100 NFP, exposed test; R: CREPE 751 FPQ / 2253 TPQ.
Gate rows route saved Plain/correction answers; not exact Two Axes reproduction.
Ours is not fixed to SAE. Source counts and comparison limits: result_sources.md.